Lei Liu

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Education

Wuhan University, China **2014.09-2020.06**
M.S. and **Ph.D.** in Geodesy and Geomatics Advisor: Dr. Yibin Yao

- Selected Coursework: Space Geodesy, Advanced Satellites Geodesy, Satellite Orbit Determination, Theory and Method of Satellite Navigation and Positioning, Optimal Estimation

Central South University of Forestry and Technology, China **2010.09-2014.06**
B.S. in Surveying and Mapping

- Selected Coursework: Principle and Application of Digital Mapping, Surveying for Engineering, GPS Surveying and Data Processing, Error Theory and Foundation of Surveying Adjustment, Advanced Mathematics

Technical Skills

- **Program Languages:** C/C++/C#, Python(Pytorch, Tensoflow, Keras, Scikit-learn, Pandas).
- **Machine Learning Models:** SVM/SVR, decision tree, MLP, CNN, RNN/LSTM, transformer, etc.
- **GNSS Algorithm and Software:** SPP/PPP/RTK; RTKLIB/GLAB/GPSTK
- **Software&Tool:** AutoCAD, ArcGIS, STK/GMAT

Professional Experiences

University of Michigan Ann Arbor, US **2025.08-Present**
Research Fellow at Department of Climate and Space Sciences and Engineering Advisor: Dr. Shasha Zou

- **Data fusion from GNSS/LEO satellites to improve ionosphere specification and support PNT service**
  * Data fusion using GNSS/LEO satellite observations to improve 3D ionospheric specification and forecasting, particularly during geomagnetic storms and solar-flare events.
  * Quantitative evaluation of GNSS Positioning, Navigation, and Timing (PNT) performance using improved ionospheric model corrections in challenging environments.

University of Colorado Boulder, US **2022.02-2025.08**
Research Associate/Affiliate at Aerospace Engineering Sciences Department Advisor: Dr. Jade Morton

- **NOAA CDP: NOAA Space Weather Data Pilot Program**
  * **The proposed effort** will analyze Spire scintillation indices in conjunction with measurements obtained by COSMIC-2 RO measurements and ground-based GNSS scintillation receivers.
- **NASA: GeoOptics GNSS Data for Space Weather Science and Beyond**
  * **Our overarching goal** is to quantify the accuracy and reliability of GeoOptics GNSS radio occultation data for space weather and its impacts on atmospheric retrievals.
- **NASA CSDA: Spire GNSS Radio Occultation (RO) Measurements Analysis**
  * **The objective** is to provide a comprehensive assessment of global ionospheric electron density measurements from Spire GNSS RO CubeSat receivers under geomagnetically disturbed days.
  * There are **strong agreements** for collocated observations between Spire RO and other data sources (IRI2016, ionosonde, ISR and COSMIC2). This demonstrates the **feasibility of using low-cost Spire CubeSats** to support ionospheric monitoring.
  * **My effort** involves data collection, pre-processing, and analysis, and performance analysis.
- **DARPA SEE: Machine Learning Prediction of Global Ionospheric TEC Maps**
  * This project extends our previous work [Liu et al. (2020)] to implement the convolutional long short-term memory **(convLSTM)** machine learning algorithm to forecast global TEC maps with **a lead time up to 24 hours**.
  * **Main contributions** include the **improved time resolution**, different **data segmentation** for training, testing, and validation, as well as the utilization of **residual prediction strategy, batch normalization, and dropout**.
  * It **outperforms the state-of-the-art method** under various levels of solar and geomagnetic activities.

University of Colorado Boulder, US **2020.08-2022.01**
Post-Doc Researcher at Aerospace Engineering Sciences Department Advisor: Dr. Jade Morton

- **DARPA SEE: Machine Learning Prediction of Storm-Time Ionospheric Irregularities**
  * **The objective** is to predict **storm-time** ionospheric **irregularities** from GPS-derived ROTI maps using a **convLSTM-based encoder-decoder** machine learning architecture.

    * **One important contribution** is the utilization of a **custom-designed loss function** that includes a **dynamic penalty** on the difference between truth and the predicted ones.
    * It can forecast **spatial structures** of the irregularities **more accurately** than those that implemented **conventional L1 and L2 loss functions**. The improvement is particularly **evident with up to a 30-min lead time**.
- **DARPA SEE: Simulation of Arctic TEC Mapping Using Integrated LEO-based GNSS-R and Ground-based GNSS Observations**
    * **The objective** is to develop a method that can construct vertical TEC (vTEC) maps based on coherent GNSS-R slant TEC (sTEC) estimations and available ground-based GNSS data, and to conduct a trade study.
    * One outstanding **technical challenge** is that GNSS-R sTEC estimations include contributions from the incident and reflection ray paths. **The spherical harmonic model** we developed that can separate them very well.
    * **Simulation results** show that the inclusion of GNSS-R measurements **improves the accuracy of the vTEC maps**.

University of Michigan Ann Arbor, US **2018.11-2020.08**
Visiting Ph.D. Student at Department of Climate and Space Sciences and Engineering Advisor: Dr. Shasha Zou

- **Forecasting Global Ionospheric TEC Using Deep Learning Approach**
    * **The objective** is to forecast global ionospheric TEC **ahead of two hours** using the **(LSTM)** machine learning network.
    * By incorporating solar and geomagnetic indices, the developed model shows **competitive performance** under various geomagnetic conditions when compared to the persistence model and two empirical models (IRI-2016 and NeQuick-2).
    * **My effort** involves data crawling, labeling, algorithm deployment, performance comparison, and analysis.
- **Multi-scale Ionosphere Responses to the May 2017 Magnetic Storm Over the Asian Sector**
    * We investigated multi-scale ionospheric responses to this event over the Asian sector by using ground-based GNSS network and spaceborne COSMIC, FY-3C RO and Swarm missions.
    * Two positive storm phases and one negative phase were identified during this event.
    * A band-like TEC enhancement was observed aligning in the northwest–southeast direction and propagated slowly southwestward near midnight during the recovery phase.

Selected journal papers

[1] **L. Liu**, Y. J. Morton, B. Breitsch, D. Hunt, R. Notarpietro, and V. Nguyen, "Estimation and evaluation of differential code bias from spire global cubesat precise orbit gnss observations", *Journal of Geodesy (under revision)*, 2026.

[2] H. Zhang, W. Luo, M. Peng, J. Kong, **L. Liu**, Y. Yao, C. Xu, F. Tang, and J. Pan, "Cascaded transformer-lstm architecture for urban canyon navigation with factor graph optimization", *GPS Solutions*, vol. 30, no. 1, p. 55, 2026.

[3] S. Zou, **L. Liu**, H. Liu, A. Coster, R. Kerr, J. Souza, and A. Santos, "Rapid evolution of super equatorial plasma bubbles and x-pattern eia formation during extreme storm conditions on 12 november 2025", *Geophysical Research Letters*, vol. 53, no. 6, e2025GL121182, 2026.

[4] P.-H. Cheng, Y. Wang, **L. Liu**, and Y. J. Morton, "Detection of traveling ionospheric disturbances triggered by 2022 tonga volcanic eruptions through cubesats coherent gnss-reflectometry measurement", *Journal of Geophysical Research: Space Physics*, vol. 129, no. 5, e2023JA032229, 2024.

[5] **L. Liu** and Y. J. Morton, "Assessment of storm-time ionospheric electron density measurements from spire global cubesat gnss radio occultation constellation", *GPS Solutions*, vol. 27, no. 2, p. 75, 2023.

[6] **L. Liu**, Y. J. Morton, P.-H. Cheng, A. Amores, C. J. Wright, and L. Hoffmann, "Concentric traveling ionospheric disturbances (ctid) triggered by the 2022 tonga volcanic eruption", *Journal of Geophysical Research: Space Physics*, e2022JA030656, 2023.

[7] **L. Liu**, Y. J. Morton, and Y. Liu, "ML Prediction of Daily Global Ionospheric TEC Maps", *Space Weather*, e2022SW003135, 2022.

[8] **L. Liu**, Y. J. Morton, and Y. Liu, "Machine Learning Prediction of Storm-time High-latitude Ionospheric Irregularities from GNSS-derived ROTI Maps", *Geophysical Research Letters*, e2021GL095561, 2021.

[9] **L. Liu**, Y. J. Morton, Y. Wang, and K.-B. Wu, "Arctic TEC Mapping Using Integrated LEO-Based GNSS-R and Ground-Based GNSS Observations: A Simulation Study", *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–10, 2021.

[10] Z. Wang, S. Zou, **L. Liu**, J. Ren, and E. Aa, "Hemispheric Asymmetries in the Mid-latitude Ionosphere During the September 7–8, 2017 Storm: Multi-instrument Observations", *Journal of Geophysical Research: Space Physics*, vol. 126, no. 4, e2020JA028829, 2021.

[11] E. Aa, S.-R. Zhang, P. J. Erickson, L. P. Goncharenko, A. J. Coster, O. F. Jonah, J. Lei, F. Huang, T. Dang, and **L. Liu**, "Coordinated Ground-Based and Space-Borne Observations of Ionospheric Response to the Annular Solar Eclipse on 26 December 2019", *Journal of Geophysical Research: Space Physics*, vol. 125, no. 11, e2020JA028296, 2020.

[12] M. Chen, **L. Liu**, C. Xu, and Y. Wang, "Improved IRI-2016 model based on BeiDou GEO TEC ingestion across China", *GPS Solutions*, vol. 24, no. 1, pp. 1–11, 2020.

[13] **L. Liu**, Y. Yao, and E. Aa, "Multi-Instrumental Observations of Early Morning Equatorial Plasma Depletions During the 2017 Memorial Weekend Storm", *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 13, pp. 5351–5357, 2020.

[14] **L. Liu**, S. Zou, Y. Yao, and E. Aa, "Multi-scale ionosphere responses to the May 2017 magnetic storm over the Asian sector", *GPS Solutions*, vol. 24, no. 1, pp. 1–15, 2020.

[15] **L. Liu**, S. Zou, Y. Yao, and Z. Wang, "Forecasting global ionospheric TEC using deep learning approach", *Space Weather*, vol. 18, no. 11, e2020SW002501, 2020.

[16] Y. Yao, X. Chen, J. Kong, C. Zhou, **L. Liu**, L. Shan, and Z. Guo, "An updated experimental model of ig indices over the antarctic region via the assimilation of iri2016 with gnss tec", *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 2, pp. 1700–1717, 2020.

[17] **L. Liu**, Y. Yao, S. Zou, J. Kong, L. Shan, C. Zhai, C. Zhao, and Y. Wang, "Ingestion of GIM-derived TEC data for updating IRI-2016 driven by effective IG indices over the European region", *Journal of Geodesy*, vol. 93, no. 10, pp. 1911–1930, 2019.

[18] J. Kong, Y. Yao, C. Zhou, Y. Liu, C. Zhai, Z. Wang, and **L. Liu**, "Tridimensional reconstruction of the Co-Seismic Ionospheric Disturbance around the time of 2015 Nepal earthquake", *Journal of Geodesy*, vol. 92, no. 11, pp. 1255–1266, 2018.

[19] **L. Liu**, Y. Yao, J. Kong, and L. Shan, "Plasmaspheric electron content inferred from residuals between GNSS-derived and TOPEX/JASON vertical TEC data", *Remote Sensing*, vol. 10, no. 4, p. 621, 2018.

[20] Y. Yao, **L. Liu**, J. Kong, and C. Zhai, "Global ionospheric modeling based on multi-GNSS, satellite altimetry, and Formosat-3/COSMIC data", *GPS Solutions*, vol. 22, no. 4, pp. 1–12, 2018.

[21] J. Kong, Y. Yao, Y. Xu, C. Kuo, L. Zhang, **L. Liu**, and C. Zhai, "A clear link connecting the troposphere and ionosphere: Ionospheric reponses to the 2015 Typhoon Dujuan", *Journal of Geodesy*, vol. 91, no. 9, pp. 1087–1097, 2017.

[22] Y. Yao, **L. Liu**, J. Kong, and F. Xinying, "Estimation of BDS DCB combining GIM and different Zero-mean constraints", *Acta Geodaetica et Cartographica Sinica*, vol. 46, no. 2, p. 135, 2017.

[23] Y. Yao, C. Zhai, J. Kong, and **L. Liu**, "Contribution of solar radiation and geomagnetic activity to global structure of 27-day variation of ionosphere", *Journal of Geodesy*, vol. 91, no. 11, pp. 1299–1311, 2017.

[24] J. Kong, Y. Yao, **L. Liu**, C. Zhai, and Z. Wang, "A new computerized ionosphere tomography model using the mapping function and an application to the study of seismic-ionosphere disturbance", *Journal of Geodesy*, vol. 90, no. 8, pp. 741–755, 2016.

[25] Y. Yao, **L. Liu**, J. Kong, and C. Zhai, "Analysis of the global ionospheric disturbances of the March 2015 great storm", *Journal of Geophysical Research: Space Physics*, vol. 121, no. 12, pp. 12–157, 2016.

[26] Y. Yao, C. Zhai, J. Kong, and **L. Liu**, "The pre-earthquake ionosphere anomaly of the 2015 nepal earthquake", *Acta Geodaetica et Cartographica Sinica*, vol. 45, no. 4, pp. 385–395, 2016.

Selected presentations

1. **L. Liu**, Y. J. Morton, P-H. Cheng, A. Amores, C. Wright, and L. Hoffmann "Concentric Traveling Ionospheric Disturbances (CTID) Triggered by the 2022 Tonga Volcanic Eruption." *Cedar Workshop 2023, San Diego, CA, US, June 25-30, 2023.* **(talk)**

2. **L. Liu**, Y. J. Morton, and Y. Liu "Machine Learning Prediction of Global Ionospheric TEC and High-latitude ROTI Maps." *Machine Intelligence for Space Applications (MISA) Seminar Series, Hosted by CU's Space Weather Technology, Research, and Education Center (SWx TREC) ,on May 3rd, 2022.* **(talk)**

3. **L. Liu**, Y. J. Morton, and Y. Wang "Arctic TEC Mapping Using Integrated GNSS-R and ground-based GNSS Observations." *In AGU Fall Meeting 2021. AGU, 2021.* **(poster)**

4. **L. Liu**, Y. J. Morton, and Y. Liu "Machine learning for Ionospheric TEC and ROTI Forecasting." *In AGU Fall Meeting 2021. AGU, 2021.* **(talk)**

5. **L. Liu**, Y. J. Morton, and Y. Liu, "Machine Learning Prediction of High-latitude Ionospheric Irregularities from GNSS-derived ROTI Maps." *In Proceedings of the 34th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2021), pp. 3870-3877. 2021.* **(talk)**

6. **L. Liu**, S. Zou, Y. Yao, and Z Wang, "Forecast of Ionosphere GNSS TEC using CNN and LSTM neural network." *In AGU Fall Meeting Abstracts, vol. 2019, pp. NG31A-0856. 2019.* **(poster)**

Projects

- NASA CDP: Spire GNSS RO Data for Ionospheric Scintillation Investigation **(participant)** 2023-present
- NASA: GeoOptics GNSS Data for Space Weather Science and Beyond **(Co-I)** 2022-2024
- NASA CSDA: Spire GNSS Radio Occultation Measurement Analysis **(participant)** 2021-2023
- DARPA: Heliosphere to Earth Atmosphere Rendering Through Building Excellent Artificial-intelligence Training **(participant)** 2020-2021
- NASA: Utilizing GNSS-R Measurements for High-latitude Ionospheric TEC Mapping **(participant)** 2021-2023

Community Service

- **Editorial Board Member** in Journal GPS Solutions since 2022.
- **Associate Editor** for AGU Journal Space Weather since 2024.
- **Guest Editor** for special issue "GNSS Ionosphere Remote Sensing: Modelling, Monitoring, Mitigation and Innovative Applications" in journal IEEE JSTARS.
- **Journal Reviewers:** Journal of Geodesy; Geophysical Research Letters; GPS Solutions; IEEE TGRS; JGR-Space Physics; Space Weather; Annales Geophysicae; IEEE JSTARS; Navigation; Remote Sensing; Frontiers in Astronomy and Space Sciences; IEEE Transactions on Plasma Science; Journal of Applied Geodesy; Advances in Space Research.
- **Grant Reviewer:** National Science Centre Poland in 2022.
- **Co-chair** for Session "Atmospheric Effects on GNSS and LEO-PNT Systems" at ION GNSS+ 2025, Baltimore, Maryland, US.
- **Co-chair** for Session "Modeling and Mitigation Techniques for Ionospheric Disturbance Effects on GNSS Positioning" at Asia Oceania Geosciences Society (AOGS) AOGS 2025, Singapore.
- **Co-chair** for Session "deep learning for GNSS remote sensing" at IEEE IGARSS 2023, Pasadena, California, US.
- **Vice-chair** for study group "AI for GNSS Remote Sensing" in the Global Geodetic Observing System (GGOS) Focus Area on "Artificial Intelligence for Geodesy –AI4G".
- **Member** for International Association Geodesy (IAG) Working Group JWG 4.3.7 "Machine Learning for Atmosphere".
- **Teaching Assistant** for the course *GNSS for Remote Sensing (ASEN6519)*, University of Colorado Boulder.
  - Supervised students on processing ground-based GPS measurements and inferring TEC and ROTI parameters.
- **Teaching Assistant** for the course *Error Theory and Foundation of Surveying Adjustment*, Wuhan University China.
  - Guided students through foundational principles of error propagation, least-squares adjustment, and surveying methodologies.

Scholarships and Awards

- National Scholarship for Doctoral Students, Wuhan University 2018 & 2017
- HI-TARGET Outstanding Student Scholarship, Wuhan University 2016
- Outstanding Graduate Student ,Wuhan University 2015
- National Scholarship, Central South University of Forestry and Technology 2014
- Outstanding Graduates of Hunan Province 2014
- National Encouragement Scholarship, Central South University of Forestry and Technology 2012 & 2011